

WEEK-2

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Use a wait() system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
root@LAPTOP-RJFD8EBT:~# nano command_executor.c
root@LAPTOP-RJFD8EBT:~# gcc command_executor.c -o command_executor
root@LAPTOP-RJFD8EBT:~# ls -l command_executor
-rwxr-xr-x 1 root root 16576 Aug 17 05:59 command_executor
root@LAPTOP-RJFD8EBT:~# ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 1449
Child PID  : 1450

Child Process
Child PID  : 1450
Parent PID : 1449

Executing command: ls
File1.txt      command_executor  task1      task3      task5      week3
"File1.txt'"  command_executor.c task1.c    task3.c    task5.c
File2.txt      process           task2      task4      task5_old
OSSP           process.c         task2.c    task4.c    tasks1

Child process completed.
root@LAPTOP-RJFD8EBT:~#
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```

root@LAPTOP-RJFD8EBT:~# uname
Linux
root@LAPTOP-RJFD8EBT:~# uname -a
Linux LAPTOP-RJFD8EBT 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
root@LAPTOP-RJFD8EBT:~# uname -r
6.18.33.2-microsoft-standard-WSL2
root@LAPTOP-RJFD8EBT:~# uname -m
x86_64
root@LAPTOP-RJFD8EBT:~# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                 8
On-line CPU(s) list:   0-7
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 3 7320U with Radeon Graphics
CPU family:             23
Model:                 160
Thread(s) per core:    2
Core(s) per socket:    4
Socket(s):              1
Stepping:               0
BogoMIPS:               4790.65
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc
                        p lm constant_tsc rep_good nopl xtopology tsc_r
                        eliable nonstop_tsc cpuid extd_apicid tsc_known
                        _freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_
                        2 movbe popcnt aes xsave avx f16c rdrand hyperv
                        isor lahf_lm cmp_legacy svm cr8_legacy abm sse4
                        a misalignsse 3dnowprefetch osvw topoext perfct
                        r_core ssbd ibrs ibpb stibp vmmcall fsgsbase bm
                        il avx2 smep bmi2 rdseed adx smap clflushopt cl
                        wb sha_ni xsaveopt xsavec xgetbv1 clzero xsavee
                        rptr arat npt nrtp_save tsc_scale vmcb_clean fl
                        ushbyasid decodeassists pausefilter pfthreshold
                        v_vmsave_vmload umip rdpid
Virtualization features:
Virtualization:         AMD-V

```

```

Virtualization features:
Virtualization:         AMD-V
Hypervisor vendor:     Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                   128 KiB (4 instances)
L1i:                   128 KiB (4 instances)
L2:                    2 MiB (4 instances)
L3:                    4 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0-7
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Old microcode:         Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; untrained return thunk; SMT enabled
                        with STIBP protection
Spec rstack overflow:   Vulnerable: Safe RET, no microcode
Spec store bypass:     Mitigation; Speculative Store Bypass disabled v
                        ia prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user
                        pointer sanitization
Spectre v2:            Mitigation; Retpolines; IBPB conditional; STIBP
                        always-on; RSB filling; PBRSE-eIBRS Not affect
                        ed; BHI Not affected
Srbds:                 Not affected
Tsa:                   Not affected
Tsx async abort:       Not affected
Vmscape:               Not affected
root@LAPTOP-RJFD8EBT:~# ps
  PID TTY          TIME CMD
 1308 pts/0    00:00:00 bash

```

```

PID TTY          TIME CMD
1308 pts/0      00:00:00 bash
1487 pts/0      00:00:00 ps
root@LAPTOP-RJFD8EBT:~# ps -ef
UID          PID    PPID  C   STIME TTY          TIME CMD
root          1        0  0  04:55 ?        00:00:02 /sbin/init
root          2        1  0  04:55 hvc0    00:00:00 /init
root          7        2  0  04:55 hvc0    00:00:00 plan9 --control-socket 7
root         59        1  0  04:55 ?        00:00:02 /usr/lib/systemd/systemd
root        104        1  0  04:55 ?        00:00:01 /usr/lib/systemd/systemd
systemd+    118        1  0  04:55 ?        00:00:00 /usr/lib/systemd/systemd
systemd+    122        1  0  04:55 ?        00:00:00 /usr/lib/systemd/systemd
root        169        1  0  04:55 ?        00:00:00 /usr/sbin/cron -f -P
message+    170        1  0  04:55 ?        00:00:00 @dbus-daemon --system --
root        186        1  0  04:55 ?        00:00:00 /usr/lib/systemd/systemd
root        195        1  0  04:55 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog      218        1  0  04:55 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root        222        1  0  04:55 tty1    00:00:00 /sbin/agetty -o -p -- \u
root        239        1  0  04:55 ?        00:00:00 /usr/bin/python3 /usr/sh
root        366        2  0  04:55 pts/1   00:00:00 /bin/login -f
root        428        1  0  04:55 ?        00:00:00 /usr/lib/systemd/systemd
root        429        428  0  04:55 ?        00:00:00 (sd-pam)
root        461        366  0  04:56 pts/1   00:00:00 -bash
polkitd     1105       1  0  05:30 ?        00:00:00 /usr/lib/polkit-1/polkit
root        1300       2  0  05:40 ?        00:00:00 /init
root        1302     1300  0  05:40 ?        00:00:01 /init
root        1308     1302  0  05:40 pts/0    00:00:00 -bash
root        1488     1308  99  06:06 pts/0    00:00:00 ps -ef

```

```

root@LAPTOP-RJFD8EBT:~# ps aux
USER          PID  %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          1  0.0  0.3  21740 13152 ?        Ss   04:55   0:02 /sbin/ini
root          2  0.0  0.0   3180  2204 hvc0    Sl+  04:55   0:00 /init
root          7  0.0  0.0   3196  2128 hvc0    Sl+  04:55   0:00 plan9 --c
root         59  0.0  0.4  50364 15932 ?        S<s  04:55   0:02 /usr/lib/
root        104  0.0  0.1  25644  7024 ?        Ss   04:55   0:01 /usr/lib/
systemd+    118  0.0  0.3  21344 13152 ?        Ss   04:55   0:00 /usr/lib/
systemd+    122  0.0  0.2  91036  7956 ?        Ssl  04:55   0:00 /usr/lib/
root        169  0.0  0.0   4244  2680 ?        Ss   04:55   0:00 /usr/sbin
message+    170  0.0  0.1   9660  5392 ?        Ss   04:55   0:00 @dbus-dae
root        186  0.0  0.2  17980  8844 ?        Ss   04:55   0:00 /usr/lib/
root        195  0.0  0.3 1756104 11068 ?        Ssl  04:55   0:00 /usr/Libe

```

```

root        428  0.0  0.3  20120 11388 ?        Ss   04:55   0:00 /usr/lib/
root        429  0.0  0.0  21172  3508 ?        S    04:55   0:00 (sd-pam)
root        461  0.0  0.1   6080  5304 pts/1    S+   04:56   0:00 -bash
polkitd     1105  0.0  0.2  308172  8048 ?        Ssl  05:30   0:00 /usr/lib/
root        1300  0.0  0.0   3184  1116 ?        Ss   05:40   0:00 /init
root        1302  0.0  0.0   3200  1252 ?        S    05:40   0:01 /init
root        1308  0.0  0.1   6076  5392 pts/0    Ss   05:40   0:00 -bash
root        1489  0.0  0.1   8288  4296 pts/0    R+   06:06   0:00 ps aux

```

```

root@LAPTOP-RJFD8EBT:~# top
top - 06:06:49 up 1:11, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem : 3549.6 total, 2883.2 free, 473.0 used, 268.4 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used, 3076.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21740	13152	9736	S	0.0	0.4	0:02.51
2	root	20	0	3180	2204	2072	S	0.0	0.1	0:00.07
7	root	20	0	3196	2128	2008	S	0.0	0.1	0:00.00
59	root	19	-1	50364	15932	14752	S	0.0	0.4	0:02.26
104	root	20	0	25644	7024	5172	S	0.0	0.2	0:01.75
118	systemd+	20	0	21344	13152	11028	S	0.0	0.4	0:00.39
122	systemd+	20	0	91036	7956	6992	S	0.0	0.2	0:00.56
169	root	20	0	4244	2680	2432	S	0.0	0.1	0:00.17
170	message+	20	0	9660	5392	4668	S	0.0	0.1	0:00.39
186	root	20	0	17980	8844	7800	S	0.0	0.2	0:00.37
195	root	20	0	1756104	11068	8536	S	0.0	0.3	0:00.82
218	syslog	20	0	222516	6000	4588	S	0.0	0.2	0:00.50
222	root	20	0	3124	1972	1836	S	0.0	0.1	0:00.01
239	root	20	0	107032	23256	13800	S	0.0	0.6	0:00.40
366	root	20	0	6676	4504	3732	S	0.0	0.1	0:00.04
428	root	20	0	20120	11388	9452	S	0.0	0.3	0:00.48
429	root	20	0	21172	3508	1796	S	0.0	0.1	0:00.00
461	root	20	0	6080	5304	3672	S	0.0	0.1	0:00.08
1105	polkitd	20	0	308172	8048	7152	S	0.0	0.2	0:00.17
1300	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
1302	root	20	0	3200	1252	1108	S	0.0	0.0	0:01.09
1308	root	20	0	6076	5392	3696	S	0.0	0.1	0:00.16
1492	root	20	0	9300	5688	3500	R	0.0	0.2	0:00.01